

Modern Python Project Management for Researchers with uv.

Elio Balestrieri

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Who am I

- Born in Genoa 
- Grew up in San Marino 
- PhD & Postdoc in Neuroscience  at University of Münster 
- HPC System Admin  at University of Münster

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Enthusiastic about many things... Relevant here: **good, open scientific software** and **reproducibility** of scientific workflows.

Who are you?

And if you feel like:

- What do you use python for?
- Why are you curious about a new package manager?

What is uv?

An extremely fast package and project manager, written in Rust

 **And why should I care?**

And why should I care?

- Extremely fast
- Very versatile for both new and existing projects
- It feels like “this is how it was supposed to be done”

OK, where's the catch?

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There is one, depending on your standpoint on Big Tech:

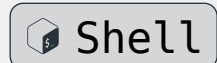
As of March 19th 2026, **Astral** (the company behind uv) **joined**
OpenAI 

<https://astral.sh/blog/openai>

 **Naja, what can we do about it?**

Not much... Decide whether to use it or not, I guess. And if you still want to do that, then let's start!

```
git clone https://github.com/ElioBalestrieri/python-uv-workshop.git
```



Installation

<https://docs.astral.sh/uv/#installation>

🍏 & 🐧:

```
$ curl -LsSf https://astral.sh/uv/install.sh | sh
```

 Shell

🪟:

```
PS> powershell -ExecutionPolicy ByPass -c "irm  
https://astral.sh/uv/install.ps1 | iex"
```

 PowerShell

Apologetic note: I have stopped using windows back in 2010 🥵

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How is everybody doing?

About virtual environments

By 🖐️: How many of you are familiar with the concept of virtual environment?

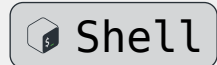
A virtual environment is:

- Contained in a directory (e.g. `.venv`), and consists of a specific Python interpreter and software libraries and binaries which are needed to support a project.
- **Isolated** from **other virtual environments** and Python interpreters and libraries installed system-wide.
- **Not** checked into source control systems such as **Git**.
- Considered as **disposable**.
- **Not** considered as **movable** or **copyable**.

uv venv / pip

You can create a new environment, with a specific python version as follows:

```
uv venv --python 3.13 my-test-venv
```



Defaults:

- no virtual env name ➞ .venv
- if python version unavailable ➞ uv downloads it for you
- no python version specified ➞ system default

uv venv / pip

You can activate the new environment as follows:

🍏 & 🐧:

```
source .venv/bin/activate
```

 Shell

🪟:

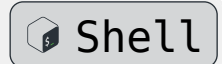
```
.venv\Scripts\activate
```

 PowerShell

uv venv / pip

You can install new packages in your environment using pip:

```
uv pip install numpy pandas
```



uv venv / pip

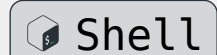
You can install new packages in your environment using pip:

```
uv pip install numpy pandas
```



And you can deactivate the virtual environment with:

```
deactivate
```



uv venv / pip

To summarize:

```
uv venv --python 3.13 my-test-venv
```

```
source my-test-venv/bin/activate
```

```
uv pip install numpy pandas
```

```
deactivate
```



Your turn 🖐️

Create a virtual environment with `uv venv` and install your favorite packages with `uv pip`

Stuck? Check: <https://docs.astral.sh/uv/pip/>

  **Why do we even need `uv` for this?**

Is this really only about having a fast `pip`?

Projects in uv

uv allows the creation of Python projects as organized development environments... Quite in a smooth way:

```
uv init my-amazing-project
```



```
cd my-amazing project
```

or, alternatively:

```
mkdir my-amazing-project
```



```
cd my-amazing project
```

```
init
```

Projects in uv

Already some interesting things:

- by default under version control with **git**
- `pyproject.toml` ➦ metadata and dependencies.

Projects in uv

```
uv run main.py
```

 Shell

Adds some items in the project:

- .venv/
- uv.lock ➔ a cross platform lockfile containing the exact info about your project dependencies. **Do not edit**

Projects in uv

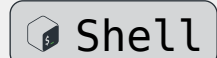
To add new dependencies:

```
uv add numpy
```



Or alternatively, edit the dependencies list of `pyproject.toml` and run:

```
uv run main.py
```



Projects in uv

uv add is also compatible with requirements.txt

```
uv add -r requirements.txt
```



Projects in uv

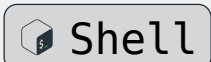
🧑 *when I was young I was simply executing my scripts with python*

OK, fine. Edit your `pyproject.toml` and then:

```
uv sync
```

```
source .venv/bin/activate
```

```
python main.py
```



Your turn 🖐️

- create a new project
- migrate the dependencies from an old project



`pip freeze > requirements.txt`

- test whether the imports work

Help yourself with these slides or:

<https://docs.astral.sh/uv/guides/projects/#managing-dependencies>

Your turn 🖐️

- create a new project
- migrate the dependencies from an old project



`pip freeze > requirements.txt`

- test whether the imports work

Help yourself with these slides or:

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alternatively, if you don't have a project to work with:

`python-uv-workshop/exercises/migrate-project`

Shipping your project

It can be as immediate¹ as:

```
uv build
```



```
uv publish --token pypi-SeCR3tT0K3N --publish-url https://test.  
pypi.org/legacy/
```

For testing your upload on test-pypi. You can even omit the `--publish-url` when you are ready to publish to PyPI.

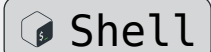
¹if you don't have C code 😇

Single executable scripts

Although it is *preferable* to work within a project, you can also declare dependencies as **inline metadata** and run the script as a standalone.

For example:

```
uv init --script my-standalone-script.py --python 3.13
```



```
uv add --script my-standalone-script.py 'numpy' 'pandas'  
'matplotlib'
```

Your turn 🖐️

Scenario

You have one python script that does not require a whole project by itself, and you want to send it around as a standalone.

How can you do that with uv?

🔄 **alternatively**, you can create a standalone script from `analysis.py` in `python-uv-workshop/exercises/migrate-project`.

Stuck? Check: <https://docs.astral.sh/uv/guides/scripts/#declaring-script-dependencies>

Useful links

<https://docs.astral.sh/uv/>

<https://pydevtools.com/>

https://www.reddit.com/r/learnpython/comments/1bmxe6i/whats_the_difference_between_pyprojecttoml/

Fin.

***Excursus:* pyproject.toml VS requirements.txt**

<https://pydevtools.com/handbook/explanation/pyproject-vs-requirements/>

requirements.txt lists packages to install. pyproject.toml defines a project. That distinction shapes how dependencies get managed and how projects scale.

Moreover requirements.txt has some gaps:

- It demands manual coordination (pip-compile or pip freeze)
- No Python version constraint.
- No project metadata.

Excursus: uv tools

Seem cool, but tbh I have never used them

<https://docs.astral.sh/uv/guides/tools/>